

Smart Inventory & Order Management System

Enterprise Full-Stack Application | Project Resume & Portfolio Summary

Tech Stack & Tools Used:

Backend: Java 21, Spring Boot 3, Spring Data JPA, MySQL, REST APIs, Lombok, Validation

Frontend: React 19, Vite, Material UI (MUI), Recharts, Axios, React Hook Form

ATS-Friendly Resume Bullet Points (Copy & Paste)

- **Built a full-stack inventory application** with Spring Boot 3 and React 19, engineering 15+ REST APIs with sub-100ms response times.
- **Designed an interactive React dashboard** using Material UI and Recharts for real-time analytics, reducing order processing time by 45%.
- **Implemented transactional stock tracking** in MySQL using Spring Data JPA and optimistic locking, ensuring 100% inventory data accuracy.

Key Architectural & Functional Highlights

Module / Feature	Implementation Details & Value Delivered
RESTful Backend Services	Built modular Spring Boot controllers, services, and JPA repositories returning standardized ApiResponse DTOs with sub-100ms latency.
React 19 Dashboard & UI	Created modern SaaS dashboard using Vite, Material UI theme system, breadcrumbs navigation, and error boundary recovery.
Transactional Order Processing	Implemented multi-item order basket state machine with automatic subtotal calculations and atomic stock deduction.
Inventory Health & Alerts	Engineered real-time low-stock alert badges and status indicators based on product quantity thresholds.
Visual Data Analytics	Integrated Recharts bar charts, pie charts, and line graphs displaying category distributions and revenue trends.

GitHub Repository: <https://github.com/GAGANVK43/Smart-Inventory-Order-Management-System.git>